CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fifth Semester of B. Tech. (CL) Examination

November 2013

CL303 Geotechnical Engineering-I

Date: 22.11.2013, Friday **Time: 10:00 a.m. To 01:00 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 (a) Enlist the some of the important applications of geotechnical engineering. [03]

(b) What are the various factors that affect the compaction of soil mass in the field? How will you measure the compaction in the field? [04]

Q - 2 (a) The following data are obtained in a Compaction test: Determine OMC & MDD. [04]

Water content (w %)	05	10	14	20	25
Bulk density (kN/m^3)	17.7	19.8	21.0	21.8	21.6

(b) Attempt Any Two;

(i) State the assumptions made in Terzaghi's theory on one dimensional consolidation. [05]

(ii) Explain square root of time fitting method to find the coefficient of consolidation. [05]

(iii) Explain theory of spring analogy for primary consolidation with neat sketches. [05]

Q - 3 (a) Define the following terms; [02]

(i) Coefficient of compressibility (ii) Coefficient of consolidation

(b) Attempt Any Two;

(i) Name and briefly explain the shear tests which may be performed based on the different drainage conditions. [06]

(ii) A stress at failure on the failure plane in a cohesionless soil mass were; normal stress = 10kN/m^2 and shear stress = 4kN/m^2. Determine the resultant stress on the failure plane, the angle of internal friction and the angle of inclination of the failure plane to the major principle plane. [06]

(iii) How will you get shear strength parameters of soil? Give limitations of Box Shear Test. [06]

SECTION – II

Q - 4 (a) Derive the following terms; [04]

(i) Water content (ii) Bulk density (iii) Void ratio (iv) Specific gravity.

(b) Give the general characteristics of the following soils; [03]

(i) Alluvial soils (ii) Aeolian soils (iii) Colluvial soils.

Q - 5 (a) (i) Write detail short note on flocculated structure with neat sketch. [02]

(ii) Enlist the uses of particle size distribution curve. [02]

(b) Attempt Any Two;

(i) The sieve analysis test was carried out for 1000 g of soil sample. The result of the sieve analysis test is shown in table below. Plot the particle size distribution curve and find the values of coefficient of curvature and uniformity coefficient. Also identify the type of soil. [05]

IS Sieve (mm)	10	4.75	2.0	1.0	0.600	0.425	0.212	0.150	0.075	Pan
Mass of soil (gm)retained	48	88	149	152	148	88	122	88	71	42

(ii) A fine-grained soil sample has the liquid limit = 40%, plastic limit = 20%. Draw plasticity chart and shown A-line in that. Then classify the soil based on Indian Standard classification system. [05]

(iii) What is liquid limit? Describe Cassagrande's to find the liquid limit. [05]

Q - 6 (a) Derive the relationship between e, G, w and Sr. [05]

(b) Write the assumptions of Darcy's law. [03]

(c) Attempt Any One;

(i) Enlist the factors affecting permeability. Explain any two in detail. [06]

(ii) A soil sample is tested for permeability and following observations are noted; Hydraulic head = 22cm, Length of specimen = 10cm, Diameter of sample = 4cm, Quality of water collected = 60cubic cm, Time taken = 15min. Determine coefficient of permeability. [06]